

Cyber Crime

Cybercrime refers to illegal/criminal activities carried out using computers, networks and the internet. With businesses, governments, and individuals relying heavily on digital platforms, cyber threats have escalated, leading to billions in financial losses worldwide. Cybercrime, especially through the Internet, has grown as the computer has become central to every field like commerce, entertainment, and government. Cybercrime may endanger a person or a nation security and financial health. Year 2023 alone has estimated over \$8 trillion in damages through cybercrime, impacting online security on a massive scale. From phishing scams to ransomware attacks, cybercriminals exploit vulnerabilities to steal sensitive information and disrupt systems. Cybercrime activities are of various categories that include:

- Identity Theft – Stealing personal information to commit fraud.
- Financial Fraud – Online scams, fake transactions, and credit card fraud.
- Cyberbullying – Harassment or threats through digital platforms.
- Phishing Attacks – Deceptive emails or websites tricking users into revealing sensitive data.
- Hacking – Unauthorized access to systems and data breaches.
- Malware Attacks – Spreading viruses, ransomware, and trojans to damage or steal data.

Cybercriminals target individuals, businesses, and even government systems, leading to significant financial losses, data breaches, and security threats.

As the internet becomes an essential part of daily life, from online shopping to business operations and communication, cybercrime cases have surged globally. Criminals exploit system vulnerabilities to steal personal data, manipulate financial transactions, and disrupt critical services

Types of Cyber Crime

Cybercrime includes a wide range of illegal activities that exploit computers, networks, and the internet. These crimes can be categorized into two main types:

- Cyber Crimes Targeting Computer Networks or Devices
- Crimes Using Computer Networks to Commit Other Criminal Activities

1) Cyber Crimes Targeting Computer Networks or Devices

These crimes involve direct attacks on computers, servers, or digital infrastructure to steal data, cause disruption, or damage systems. It involves different threats like- viruses, bugs, etc.

- **Malware Attacks:** This kind of cyber threat relates to malware viruses, worms, Trojans, etc. for interfering, damaging, or unauthorized access to computer systems.
Example: ransomware encrypts files and then later demands ransom for decryption.
- **Denial-of-Service (DoS) Attacks:** Here, the attackers focus on a system and flood it with high traffic, hence making it inaccessible to the users. Another dangerous variant of DoS is DDoS, wherein many compromised systems target one, thus, much difficult to defend against.

Example: an attack crashes an e-commerce website by overwhelming its server with traffic.

- **Phishing Attacks:** These are masqueraded e-mails or messages claiming to be from a formal web but only request that the user grant access to sensitive information like password points for an account or credit card numbers. Phishing can be described as an outstanding one of the most common cyber threats.

Example: A fake PayPal login page that steals your credentials.

- **Botnets (Zombie Networks):** A number of hijacked computers can become a "botnet" of malware that can be used by an attacker for coordinated attacks or spamming.

Example: Hackers use botnets to send millions of spam emails in a single day.

- **Exploits and Vulnerabilities:** The typical area through which cyber-thieves exploit software weakness is the application or operating system vulnerability in order to access it illegally.

Example: Exploiting an outdated banking app to steal user financial details.

2) Crimes Using Computer Networks to Commit Other Criminal Activities

These types of crimes include cyberstalking, financial fraud, or identity thief.

- **Cyberstalking:** This is considered as that crime in the nature of threatening or frightening a person on-line and spreading fear and emotional distress. This can be termed as involving threats, constant monitoring, or receiving repeated unwanted messages.

Example: Sending threatening messages to a person via email or social media.

- **Financial Fraud:** This is an example of a cybercrook manipulating the victim online to proceed with stealing money, such as fake investment opportunities, hacking a business email, and using someone else's credit card details.

Example: A fake online store that steals credit card details without delivering products.

- **Identity Theft:** It is normally the identity of people whose information is stolen with the intention of only acting like them either to misuse their cash or money from their account or even to do malicious reasons. It always lowers the credit score of the victim and in the worst case scenario, misused the account/loan financially with incorrect transactions.

Example: A hacker using stolen credentials to apply for credit cards and loans.

- **Online Harassment and Hate Crimes:** When people use the internet to discriminate against a particular person based on his or her racial background, gender, religion, or whatever, which can psychologically disturb the harassed person.

Example: Cyberbullying campaigns that target individuals based on race, gender, or religion.

- **Intellectual Property Theft:** Intellectual property theft refers to the theft of copyrighted content or business secrets via internet, financially and competitively hurting individuals and companies.

Example: A software company illegally using another firm's source code to create a competing product.

Challenges of Cyber Crime

- People are unaware of their cyber rights: The Cybercrime usually happen with illiterate people around the world who are unaware about their cyber rights implemented by the government of that particular country.
- Anonymity: Those who Commit cyber crime are anonymous for us so we cannot do anything to that person.
- Less numbers of case registered: Every country in the world faces the challenge of cyber crime and the rate of cyber crime is increasing day by day because the people who even don't register a case of cyber crime and this is major challenge for us as well as for authorities as well.
- Mostly committed by well educated people: Committing a cyber crime is not a cup of tea for every individual. The person who commits cyber crime is a very technical person so he knows how to commit the crime and not get caught by the authorities.
- No harsh punishment: In Cyber crime there is no harsh punishment in every cases. But there is harsh punishment in some cases like when somebody commits cyber terrorism in that case there is harsh punishment for that individual. But in other cases there is no harsh punishment so this factor also gives encouragement to that person who commits cyber crime.

Impact of Cyber Crimes

- Financial Losses: The fraud and theft can cause great losses not only for the given organizations but for individuals also.
- Reputational Damage: Some people may realize that reputation becomes an issue they may lose depending on the legal outcomes resulting from lawsuits.
- Operational Disruption: As will be highlighted later, such an occurrence leads to a shutdown and consequently a loss of productivity.
- Legal Consequences: In the cases where clients have been involved in some legal cases or even regulatory fines, they may have to go through another phase of legal activities, clients have to spend considerable amount of money on protecting their data.

Classification of cyber crimes

Generally, almost all cyber crimes can be classified under three heads, depending on the groups they are targetted at.

- Cyber crimes against individuals
 - Cyberbullying
 - Cyberstalking
 - Cyber defamation
 - Phishing
 - Cyber fraud
 - Cyber theft
 - Spyware
- Cyber crimes against organizations
 - Attacks by virus
 - Salami attack
 - Web Jacking
 - Denial of Service Attack
 - Data diddling
- Cyber crimes against society at large
 - Cyber pornography
 - Cyber terrorism
 - Cyber Espionage

Cyber Security

Cyber security is a potential activity by which information and other communication systems are protected from and/or defended against the unauthorized use or modification or exploitation or even theft. It is a well-designed technique to protect computers, networks, different programs, personal data, etc., from unauthorized access.

All sorts of data whether it is government, corporate, or personal need high security; however, some of the data, which belongs to the government defense system, banks, defense research and development organization, etc. are highly confidential and even small amount of negligence to these data may cause great damage to the whole nation. Therefore, such data need security at a very high level. In a larger perspective to make any security system strong, organisation need to pay attention to the following factors

- Security Architecture
- Network Diagram
- Security Assessment Procedure
- Security Policies
- Risk Management Policy
- Backup and Restore Procedures
- Disaster Recovery Plan
- Risk Assessment Procedures

Security Architecture defines the framework, Network Diagram visualizes the network, Security Assessment Procedure identifies vulnerabilities, Security Policies outline rules, Risk Management Policy manages potential threats, Backup and Restore Procedures ensure data recovery, Disaster Recovery Plan outlines incident response, and Risk Assessment Procedures identify and prioritize risks. Once a society have a complete blueprint of the points mentioned above, they can easily put better security system and can also retrieve the data if something goes wrong.

CISCO proposes a successful cybersecurity posture to have multiple layers of protection spread across the computers, networks, programs, or data that one intends to keep safe. In any organization, a unified threat management gateway system should automate integrations across products and accelerate key security operations functions like detection, investigation, and remediation. Moreover the major 3 components People, Processes, and Technology must all complement one another to create an effective defense from cyberattacks., as detailed below.

- **People** : Users must understand and comply with basic data protection and privacy security principles like choosing strong passwords, being wary of attachments in email, and backing up data.
- **Processes** : Organizations must have a framework for how they deal with both attempted and successful cyberattacks. Models like NIST cybersecurity framework, can explain how to identify attacks, protect systems, detect and respond to threats, and recover from successful attacks.
- **Technology** : Technology is essential to giving organizations and individuals the computer security tools needed to protect themselves from cyberattacks. Three main entities must be protected: endpoint devices like computers, smart devices, and routers; networks; and the cloud. Common technology used to protect these entities include next-generation firewalls,

Domain Name System (DNS) filtering, malware protection, antivirus software, and email security solutions.

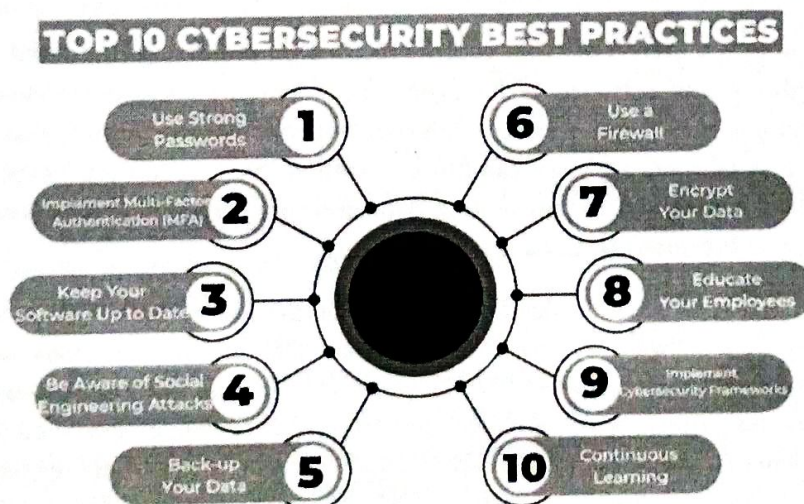
Best practices for Cybersecurity

Adopting any best practices for cybersecurity can significantly reduce the risk of cyberattacks. Listed below are three CISCO proposed significant key practices:

- *Regular software and operating system updates*
Updating software and operating systems regularly helps to patch vulnerabilities and enhance security measures against potential threats.
- *Using strong and unique passwords*
Creating strong and unique passwords for each online account can enhance cybersecurity, as cyberattacks often exploit weak or stolen passwords.
- *Implementing multi-factor authentication (MFA)*
Multi-factor authentication involves multiple identification forms before account access, reducing the risk of unauthorized access.



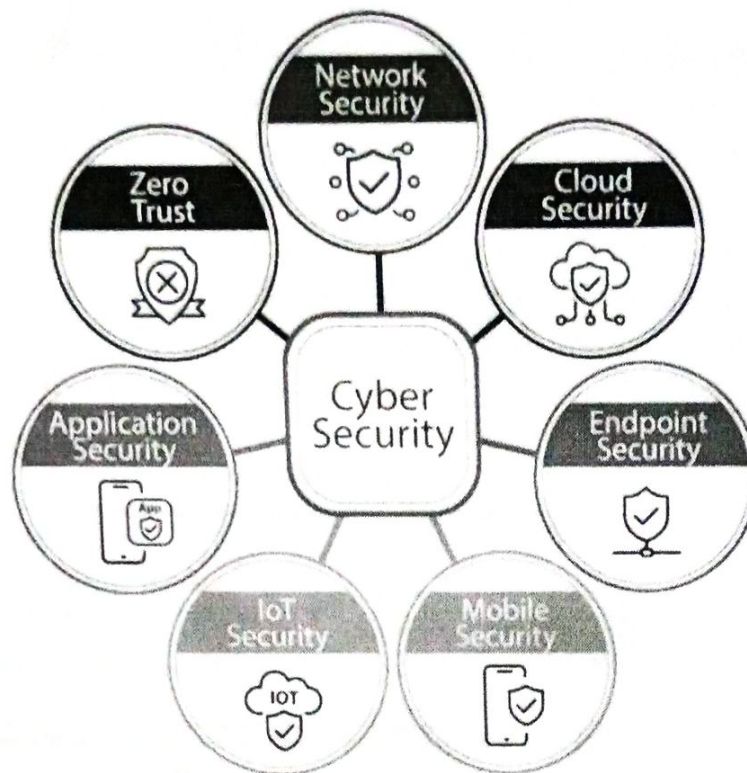
Yet another comprehensive model proposed by Clarusway School has TEN mantras as defined in picture below



Carrying out these practices enhances cybersecurity and protects digital assets. Above all It's vital for any groups/organisation to adopt Cyber Security as a Culture, so as to stay vigilant and remain informed about the latest threats and security measures thereby staying ahead of cybercriminals.

Case Study: Wide Disciplines of CyberSecurity.. Exerts from Keepnet Security WhitePaper

Cybersecurity Ventures reports that global costs from cyber attacks are expected to reach \$10.5 trillion by 2025, underlining the urgent need for improved cyber security. By strengthening cyber security measures and improving cyber threat detection, organizations can protect their assets and reduce the financial and operational impacts of these attacks.



Cyber security is a wide field covering several disciplines. It can be divided into seven main pillars:

Network Security

Network security is a branch of cyber security focused on protecting computer networks and data through hardware and software technologies. It's important because it prevents unauthorized access and misuse of network resources. Effective network security ensures that networks operate reliably and that business activities can proceed without interruption from cyber security threats. This protection is essential for preventing data breaches, protecting sensitive information, and maintaining trust between users and service providers.

Examples of Network security include Firewalls, which control traffic based on security rules; Antivirus and Anti-malware Software, which detect and remove malicious software; Intrusion Detection and Prevention Systems (IDS/IPS), which monitor and block suspicious activities; Virtual Private Networks (VPNs), which encrypt internet traffic; Network Access Control (NAC), which restricts network access to authorized devices; and Encryption, which secures data sent across the network.

Cloud Security

Cloud Security is a segment of cyber security dedicated to protecting cloud-based systems, including data, applications, and infrastructures. It plays a key role in preventing unauthorized access and data breaches, as well as managing other security threats to cloud environments. By implementing strong

cloud security measures, organizations can confidently utilize cloud services without compromising the safety of their sensitive information. Cloud Security ensures smooth operations, maintaining data privacy, and building trust between users and cloud service providers.

Examples of cloud security include Data Encryption, which secures data both stored and in transit; Access Management tools, which ensure only authorized users can access certain data or applications; Security Configuration Management, which maintains secure settings across cloud services; Intrusion Detection Systems (IDS), which monitor for malicious activities; and Security Information and Event Management (SIEM) systems, which provide real-time analysis and reporting on security alerts generated by applications and network hardware.

Endpoint Security

Endpoint security is a critical aspect of cyber security focused on protecting devices connected to company networks, such as desktops, laptops, and mobile devices. This field is essential for blocking unauthorized access and preventing breaches that target these endpoints. Effective endpoint security strategies enable organizations to control device access to networks, ensuring that sensitive data remains secure. This level of security is fundamental for ensuring uninterrupted operations, protecting confidential data, and building trust among all involved parties.

Examples of endpoint security include Antivirus Programs, which protect devices from malware and viruses; Endpoint Detection and Response (EDR), which continuously monitors and responds to security threats; Device Management Solutions, which manage and control device access to the network; Application Control, which prevents unauthorized software from operating; and Data Loss Prevention (DLP), which helps ensure sensitive data does not exit the network. These tools collectively enhance cyber security by protecting devices from a variety of cyber threats.

Mobile Security

Mobile security is a key type of cyber security that focuses on protecting mobile devices such as smartphones, tablets, and laptops from unauthorized access, data breaches, and other cyber threats. It provides cyber security protection for sensitive information stored on these devices, ensuring data accuracy and safety. Strong mobile security practices allow users to engage safely with digital services while mobile without risking data exposure. Implementing robust mobile security measures helps preserve data privacy, support secure mobile transactions, and maintain user trust.

Examples of mobile security include Mobile Device Management (MDM), which manages and secures mobile devices within an organization; Biometric Security, which uses physical characteristics like fingerprints or facial recognition to authenticate users; Secure Wi-Fi, which employs encryption protocols to protect data sent over public networks; Mobile Antivirus Software, which detects and removes malware; and Mobile Application Management (MAM), which controls and secures applications on mobile devices. These protections are essential for defending against the growing number of mobile-specific cyber threats.

IoT Security

IoT security is a specialized area of cyber security that focuses on protecting devices like smart home heaters, fitness trackers, and smart household items from unauthorized access and cyber attacks. It is essential to protect the vast amounts of data these devices collect and transmit and prevent unauthorized control of the devices themselves. By adopting strong IoT security protocols,

organizations can protect their device ecosystems, enhancing the overall security of their networks. This high level of IoT security is vital for ensuring IoT devices operate correctly, maintaining user privacy, and securing the reliability of connected systems.

Examples of IoT security include Device Authentication, which verifies the identity of devices before allowing them to connect to networks; Secure Firmware Updates, which provide regular security enhancements to IoT devices; Network Segmentation, which isolates IoT devices in separate network zones to reduce the risk of widespread network breaches; Encryption, which secures data sent from IoT devices to other devices and servers; and Continuous Monitoring, which tracks device behavior for signs of compromise or unusual activity. These measures are key to defending against the unique threats that target IoT environments.

Application Security

Application security is a type of cyber security that focuses on protecting software applications from cyber threats and vulnerabilities. It aims to block unauthorized access and protect sensitive data within applications. With robust application security measures, organizations can confidently run their software, knowing that both their own and their users' data is secured. Application security is key to maintaining the smooth operation of applications, keeping safe data privacy, and building trust between users and developers.

Examples of application security include Static Application Security Testing (SAST), which analyzes source code for vulnerabilities; Dynamic Application Security Testing (DAST), which tests running applications to find security issues; Web Application Firewalls (WAF), which protect applications from incoming threats; Secure Coding Practices, which involve guidelines to write safer application code; and Application Patch Management, which involves regularly updating applications with security updates to address discovered vulnerabilities. These strategies are important for maintaining the stability and security of software applications against evolving cyber threats.

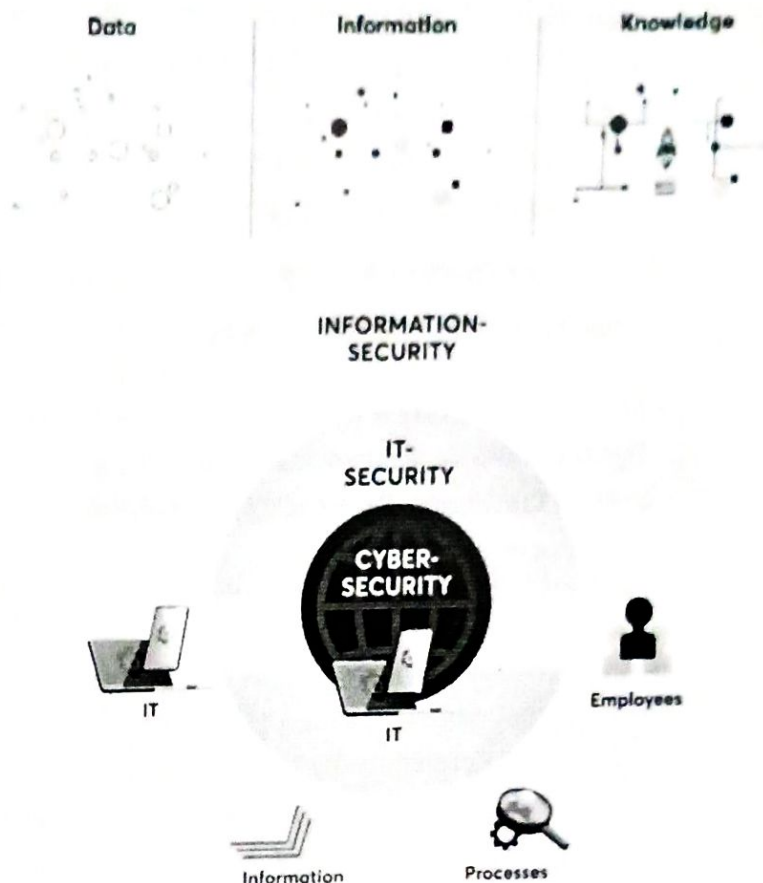
Zero Trust

Zero Trust is a cyber security strategy that operates on the principle of "never trust, always verify." This approach insists on continuously verifying all users and devices, both inside and outside the organization's network before granting access to any resources or data. By requiring strict identity verification, Zero Trust ensures that only authenticated and authorized users can access specific applications or data. This method significantly improves cybersecurity by reducing the areas that attackers can target and lowering the chance of unauthorized access.

Examples of Zero Trust are Multi-Factor Authentication (MFA), which strengthens security by requiring additional identity verification methods beyond standard passwords; Least Privilege Access Control, which restricts user access to only the essentials necessary for their roles; Microsegmentation, which divides the network into smaller sections to limit damage from breaches and stop unauthorized movements across the network; Real-Time Threat Detection, which continuously monitors activities to quickly identify and respond to suspicious behavior; and Continuous Security Monitoring, which constantly checks security events to make sure Zero Trust rules are followed. These practices are essential for the effective implementation of the Zero Trust model, providing robust protection against modern cyber security threats.

Information Security

Information security is the practice of protecting information by moderating information risks. It involves the protection of information systems and the information processed, stored, and transmitted by these systems from unauthorized access, use, disclosure, disruption, modification, or destruction. This includes the protection of personal information, financial information, and sensitive or confidential information stored in both digital and physical forms. Effective information security requires a comprehensive and multi-disciplinary approach, involving people, processes, and technology. Information Security spans so many research areas like Cryptography, Mobile Computing, Cyber Forensics, Online Social Media, etc.



During the First World War, a Multi-tier Classification System was developed keeping in mind the sensitivity of the information. With the beginning of the Second World War, formal alignment of the Classification System was done. Alan Turing was the one who successfully decrypted the Enigma Machine which was used by Germans to encrypt warfare data.

Effective information security requires a comprehensive approach that considers all aspects of the information environment, including technology, policies and procedures, and people. It also requires ongoing monitoring, assessment, and adaptation to address emerging threats and vulnerabilities.

Usage of Information Security

Information security is generally used to protect valuable information assets from a wide range of threats, including theft, espionage, and cybercrime. Some key reasons on its importance is listed

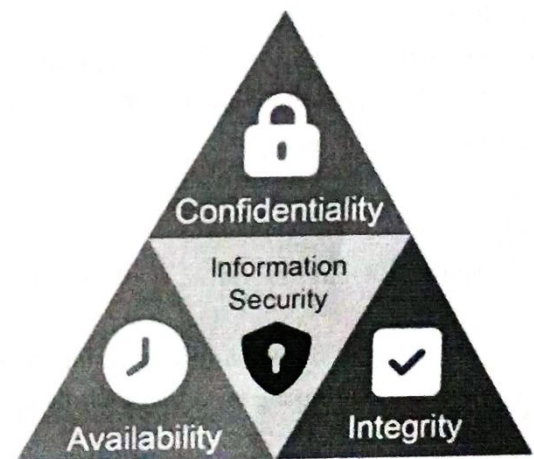
- **Protecting sensitive information:** Information security helps protect sensitive information from being accessed, disclosed, or modified by unauthorized individuals. This includes personal information, financial data, and trade secrets, as well as confidential government and military information.
- **Mitigating risk:** By implementing information security measures, organizations can mitigate the risks associated with cyber threats and other security incidents. This includes minimizing the risk of data breaches, denial-of-service attacks, and other malicious activities.
- **Compliance with regulations:** Many industries and jurisdictions have specific regulations governing the protection of sensitive information. Information security measures help ensure compliance with these regulations, reducing the risk of fines and legal liability.

- Protecting reputation: Security breaches can damage an organization's reputation and lead to lost business. Effective information security can help protect an organization's reputation by minimizing the risk of security incidents.
- Ensuring business continuity: Information security helps ensure that critical business functions can continue even in the event of a security incident. This includes maintaining access to key systems and data, and minimizing the impact of any disruptions.

3 Principles of Information Security

Information security is necessary to ensure the authentication of information, whether it is stored digitally or physically. Information Security programs are built around 3 objectives, commonly known as CIA – Confidentiality, Integrity, Availability.

- Confidentiality – Means information is not disclosed to unauthorized individuals, entities and process. For example if we say I have a password for my Gmail account but someone saw while I was doing a login into Gmail account. In that case my password has been compromised and Confidentiality has been breached.
- Integrity – Means maintaining accuracy and completeness of data. This means data cannot be edited in an unauthorized way. For example if an employee leaves an organisation then in that case data for that employee in all departments like accounts, should be updated to reflect status to JOB LEFT so that data is complete and accurate and in addition to this only authorized person should be allowed to edit employee data.
- Availability – Means information must be available when needed. For example if one needs to access information of a particular employee to check whether employee has outstanding the number of leaves, in that case it requires collaboration from different organizational teams like network operations, development operations, incident response and policy/change management. Denial of service attack is one of the factor that can hamper the availability of information.



Cyber Crimes- Tools and Methods

- **Password Cracking** : Password cracking is the process of guessing passwords[1] protecting a computer system. A common approach (brute-force attack) is to repeatedly try guesses for the password and to check them against an available cryptographic hash of the password. The purpose of password cracking might be to help a user recover a forgotten password (due to the fact that installing an entirely new password would involve System Administration privileges), to gain unauthorized access to a system, or to act as a preventive measure whereby system administrators check for easily crackable passwords. There are different types of Password cracking like Online attacks. Offline attacks or Non-electronic attacks
- **Key loggers** : Keystroke logging, often called keylogging, is the practice of noting (or logging) the keys struck on a keyboard, typically in a covert manner so that the person using the keyboard is Unaware that such action are being monitored. It can be classified as software key logger and hardware key logger.
 - **Software Key loggers**. Software keyloggers are software programs installed on the computer systems which usually are located between the OS and the keyboard hardware, and every keystroke is recorded. Software keyloggers are installed on a computer system by Trojans or viruses without the knowledge of the user.
 - **Hardware Key loggers**: Hardware keyloggers are small hardware devices. These are connected to the PC and/or to the keyboard and save every keystroke into a file or in the memory of the hardware device. Cybercriminals install such devices on ATM machines to capture ATM Cards' PINs. Each Key press on the keyboard of the ATM is registered by these key loggers.
- **Spyware** is a type of malware that is installed on computers which collects information about users without their knowledge. It is clearly understood from the term Spyware that it secretly monitors the user. The features and functions of such Spywares are beyond simple monitoring.
- **SQL injection**, also known as SQLi, is a common attack vector that uses malicious SQL code for backend database manipulation to access information that was not intended to be displayed. This information may include any number of items, including sensitive company data, user lists or private customer details. Types of SQL Injections include
 - In-band SQLi
 - Error-based SQLi
 - Union-based SQLi
 - Inferential (Blind) SQLi
 - Out-of-band SQLi
- **Network Access Control (NAC)**, also known as network admission control, is the process of restricting unauthorized users and devices from gaining access to a corporate or private network. NAC ensures that only users who are authenticated and devices that are authorized and compliant with security policies can enter the network. Advantages of Network Access Control include
 - Control the users entering the corporate network

- Control access to the applications and resources users aim to access
 - Allow contractors and guests to enter the network but restrict their access
 - Segment employees into groups and build role-based access policies
 - Protect against cyberattacks by detecting unusual or suspicious activity
 - Automate incident response
 - Generate reports and insights on attempted access across the organization
- Cloud Security: Cloud security is a responsibility that is shared between the cloud provider and the customer. There are basically three categories of responsibilities in the Shared Responsibility Model:
 - responsibilities that are always the provider's,
 - responsibilities that are always the customer's, and
 - responsibilities that vary depending on the service model,
 - Infrastructure as a Service (IaaS),
 - Platform as a Service (PaaS), or
 - Software as a Service(SaaS), such as cloud email.
- Web Security: Web Security deals with the security of data over the internet/network or web or while it is being transferred to the internet. For e.g. when you are transferring data between client and server and you have to protect that data that security of data is your web security.
- Wireless Security : Wireless security is the prevention of unauthorized access or damage to computers or data using wireless networks, which include Wi-Fi networks. The term may also refer to the protection of the wireless network itself from adversaries seeking to damage the confidentiality, integrity, or availability of the network.